SPECTRAL ASPECT SUBCONVEXITY FOR PGL_2: A COUPLE APPROACHES

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Abstract. We discuss and compare two ways to estimate a short second moment of L-functions on PGL_2 in the spectral aspect: that of Iwaniec [7], using the approximate functional equation and Kuznetsov formula, and that of [12] using period integrals.

Remark 1. This is a lightly edited version of "log-2022-03-08-a.tex". It needs a lot of polishing, but might be useful to someone even in its current state.

1. Goal

Take

$$G := \mathrm{PGL}_2(\mathbb{R}),$$
$$\Gamma := \mathrm{PGL}_2(\mathbb{Z}) \hookrightarrow G,$$
$$H := \mathrm{GL}_1(\mathbb{R}) \cong \begin{pmatrix} * & 0 \\ 0 & 1 \end{pmatrix} \hookrightarrow G,$$
$$\Gamma_H := \mathrm{GL}_1(\mathbb{Z}) = \{\pm 1\} \hookrightarrow H.$$

Let $\pi \subset L^2_{\mathrm{cusp}}(\Gamma\backslash G)$ be a cuspidal automorphic representation corresponding to a Maass form of eigenvalue $1/4 + T^2$. We aim to show that

$$L(\pi, \tfrac{1}{2}) \ll T^{1/2-\delta} \tag{1.1}$$

for some fixed $\delta > 0$.

Such an estimate was first shown by Iwaniec [7] (initially under an additional hypothesis, which Iwaniec later observed could be removed using the identity $\lambda(p)^2 - \lambda(p^2) = 1$). Iwaniec's method consists of estimating an amplified second moment over T in an interval of length a bit more than 1. This method was adapted to

PGL_3 by Blomer–Buttcane [3, 4], using an amplified fourth moment, and then to GL_n in [12] (direct link), using an amplified $2(n-1)$th moment. In the works of Iwaniec and Blomer—Buttcane, the moment is estimated using the approximate functional equation, Kuznetsov formula, and several applications of Poisson summation. In [12], the moment is estimated implicitly by direct analysis of an integral representation for the L-function, with vectors chosen as in [14] and the averaging implemented via the pretrace formula like in [8].

Remark 2. The best known bound for (1.1) is to due to Ivic [6] , who showed that (1.1) holds for any fixed $\delta < 1/6$. Ivic's method consists of estimating an unamplified fourth moment over T in an interval of length a bit more than $T^{1/3}$ (compare with [10, 11, 5, 2]). Such approaches using "higher moments" have no known extension beyond GL_2. It would be significant to identify such an extension.

2. Automorphic background

2.1. **Cuspidal automorphic representations.** The space $L^2_{\mathrm{cusp}}(\Gamma\backslash G)$ consists of square-integrable functions on $\Gamma\backslash G$ whose constant term vanishes. It is simultaneously a representation for the group G, acting via right translation, and the Hecke algebra, acting via left translation with respect to double cosets in $\Gamma\backslash \mathrm{PGL}_2(\mathbb{Q})/\Gamma$. Under these actions, it decomposes as a direct sum of irreducible representations, each occurring with multiplicity one. We denote by π the space of smooth vectors inside one such representation.

2.2. **Whittaker expansion.** The Hecke eigenvalues of π are described by a multiplicative function $\lambda_\pi : \mathbb{N} \to \mathbb{C}$, which specifies the eigenvalues for a spanning set of double cosets. For $\varphi \in \pi$, one defines the Whittaker function $W_\varphi : G \to \mathbb{C}$ by

$$W_\varphi(g) := \int_{x \in \mathbb{R}/\mathbb{Z}} \varphi\left(\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} g\right) e(-x)\, dx,$$

where we employ the standard abbreviation $e(t) := e^{2\pi i t}$. Conversely, φ may be recovered from λ_π and W_φ via the formula

$$\varphi(g) = \sum_{n \neq 0} \frac{\lambda_\pi(|n|)}{|n|^{1/2}} W_\varphi\left(\begin{pmatrix} n & 0 \\ 0 & 1 \end{pmatrix} g\right), \tag{2.1}$$

with the sum taken over all nonzero integers n.

2.3. **Kirillov model.** The Whittaker model $\mathcal{W}(\pi)$ for π is defined to consist of all functions of the form W_φ. The theory of the Kirillov model implies that the restriction map

$$\mathcal{W}(\pi) \to \{\text{functions } H \to \mathbb{C}\}$$

is injective, and that its image contains $C_c^\infty(H)$. Combining this fact with (2.1) tells us that

- any $\varphi \in \pi$ is determined by $W_\varphi : H \to \mathbb{C}$, and
- every smooth compactly-supported function $H \to \mathbb{C}$ determines a unique $\varphi \in \pi$.

3. Integral representation

The L-function $L(\pi, s)$ may be defined for $\Re(s) > 1$ by the absolutely convergent Dirichlet series $\sum_{n\in\mathbb{N}} \lambda_\pi(n) n^{-s}$ and in general by meromorphic continuation. By unfolding (2.1), one obtains the Hecke/Jacquet–Langlands integral representation

$$\int_{\Gamma_H\backslash H} \varphi = L(\pi, \tfrac{1}{2}) \int_H W_\varphi. \tag{3.1}$$

4. Coadjoint orbits

We denote by $\mathfrak{g}$ the Lie algebra of G and by $\mathfrak{g}^*$ its linear dual. Using the trace pairing, we may identify $\mathfrak{g}^*$ with $\mathfrak{g}$, which we identify further with the space $\mathfrak{sl}_2(\mathbb{R})$ of traceless 2×2 matrices ξ. The coadjoint orbit

$$\mathcal{O}_\pi \subseteq \mathfrak{g}^*$$

attached to π is the one-sheeted hyperboloid cut out by the equation

$$\det(\xi) = -T^2.$$

(See for instance §**??**-§**??** of Quantum variance III.) It comes with a natural symplectic volume form that describes the character of π via the Kirillov formula (see [14, §6]). We note that in the coordinates

$$\xi = \begin{pmatrix} a & b \\ c & -a \end{pmatrix} = \begin{pmatrix} x & y-z \\ y+z & -x \end{pmatrix},$$

we have

$$\det(\xi) = -a^2 - bc = z^2 - y^2 - x^2.$$

In particular, $\mathcal{O}_\pi$ contains the circle $\{(x, y, 0) : x^2 + y^2 = T^2\}$. The circular strip $\{(x, y, z) \in \mathcal{O}_\pi : |z| \le 1/2\}$ has symplectic volume one; in the orbit method heuristic described in [14, §1.7], it corresponds to the weight zero vector in π.

The integrals (3.1) may be understood as describing how π oscillates against the trivial character of H. The orbit method suggests [14, §1.9] that such oscillation may be understood in terms of the intersection

$$\mathcal{O}_\pi \cap \mathfrak{h}^\perp,$$

i.e., the preimage in $\mathcal{O}_\pi$ of the trivial element 0 of $\mathfrak{h}^*$. That intersection is given in (x, y, z) coordinates by

$$\{(0, y, z) : y^2 - z^2 = T^2\}$$

and in (a, b, c) coordinates by

$$\{(0, b, c) : bc = T^2\}.$$

It is a closed H-orbit, with trivial stabilizer.

We pick a point $\tau \in \mathcal{O}_\pi \cap \mathfrak{h}^\perp$ of size comparable to T. For concreteness, let us take

$$\tau = \begin{pmatrix} 0 & T \\ T & 0 \end{pmatrix}.$$

Remark 3. For the variant problem concerning π attached to a holomorphic form of weight $2k$, we would take $T := k - 1/2$, we would take $\mathcal{O}_\pi$ cut out by $\det(\xi) = T^2$, and we would take

$$\tau = \begin{pmatrix} 0 & -T \\ T & 0 \end{pmatrix}.$$

See for instance §**??** of Quantum variance III.

5. Choice of vector

We seek a unit vector $\varphi \in \pi$ that is "localized at τ." Informally, this means that for each fixed Lie algebra element $X \in \mathfrak{g}$, we have

$$X\varphi = i\langle X, \tau\rangle\varphi + \mathrm{O}(T^{1/2}). \tag{5.1}$$

For further informal discussion of this concept, see [14, §1.7] and [13, §2.5] . For a precise definition, see §3 of this note; for exercises, see this note; for further discussion, see §**??** of Quantum variance III, or [12, §14] (direct link: §**??**).

Such a vector φ may be described readily in the Kirillov model for π. Recall (from §2.3) that this model consists of the restrictions to H of elements W of the Whittaker model $\mathcal{W}(\pi)$. It will be convenient to think of such restricted elements W as functions on $\mathbb{R}^\times$ via the abbreviation

$$W(y) := W\left(\begin{pmatrix} y & 0 \\ 0 & 1 \end{pmatrix}\right).$$

We consider the following basis elements for $\mathfrak{g}$:

$$\partial_a = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \partial_b = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \partial_c = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

The action of the first two of these elements on the Kirillov model is given very simply:

$$\partial_a W(y) = 2yW'(y), \quad \partial_b W(y) = 2\pi i y W(y).$$

From now on, we will be very vague and informal with asymptotic notation. For a precise discussion, we refer to Exercise **??** of this note. Since $\langle \partial_a, \tau\rangle = 0$ and $\langle \partial_b, \tau\rangle = T$, the condition (5.1) says in particular that

$$\partial_a W_\varphi = \mathrm{O}(T^{1/2}), \quad \partial_b W_\varphi = iTW_\varphi + \mathrm{O}(T^{1/2}).$$

These formulas suggest taking the following smoothened L^2-normalized characteristic function:

$$W_\varphi(y) = T^{1/4} 1^{\mathrm{smooth}}_{T/2\pi + \mathrm{O}(T^{1/2})}(y). \tag{5.2}$$

Using that W_φ is an eigenfunction under the Casimir operator, one can check also that

$$\partial_c W_\varphi = iTW_\varphi + \mathrm{O}(T^{1/2}).$$

This is the content of Exercise **??** of this note. For GL_n, precise forms of all these estimates are established in [12, Part 3] (direct link: Part **??**).

Remark 4. This choice of φ is closely related to the "microlocal lift" of Zelditch [17] et al., see [9, 16, 1]. More precisely, it is (asymptotically) a G-translate of the usual definition; the limit invariance for L^2-masses will be with respect to the stabilizer G_τ of τ, namely

$$G_\tau = \left\{ \begin{pmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{pmatrix} \right\}, \tag{5.3}$$

rather than with respect to the diagonal subgroup.

By choosing the smooth bump function implicit in (5.2) to be nonnegative, we may arrange that

$$\int_H W_\varphi \asymp T^{-1/4},$$

so that (3.1) reads

$$T^{-1/4}L(\pi, \tfrac{1}{2}) \asymp \int_{\Gamma_H \backslash H} \varphi. \tag{5.4}$$

Our task is to show that the right hand side is $\ll T^{1/4-\delta}$.

6. Truncation

We now indicate why the integral on the right hand side of (5.4) of may be effectively truncated to a fixed compact set. For a detailed discussion of this point, see [12, §5.3] (direct link: **??**) or [10, §5.1.4] or [15, §3] (or §3 of this informal note).

Consider, for a fixed even "truncation" function $\mathcal{T} \in C_c^\infty(H)$, the map

$$I : H \to \mathbb{C}$$

$$I(Y) := \int_{y \in \Gamma_H \backslash H} \mathcal{T}(y/Y)\varphi(y)\, d^\times y$$

assigning to a parameter Y the smoothened integral of $\varphi|_H$ over the corresponding dyadic range. We have

$$\int_{\Gamma_H \backslash H} \varphi |.|^{s-1/2} = L(\pi, s) Z(W_\varphi, s),$$

where $Z(W_\varphi, s)$ denotes the local zeta integral

$$Z(W_\varphi, s) := \int_H W_\varphi |.|^{s-1/2}.$$

For $s = \mathrm{O}(1)$, we see by explicit calculation that

$$Z(W_\varphi, s) \approx T^{s-3/4}.$$

If moreover $|\Re(s)| \leq 1/2$, then the convexity bound reads

$$L(\pi, s) \ll T^{1-s}.$$

These bounds become only polynomially worse if we relax the condition $s = \mathrm{O}(1)$ to $\Re(s) = \mathrm{O}(1)$. Multiplying them together and convolving against the rapidly-decaying Mellin transform of the fixed test function $\mathcal{T}$, we deduce the Mellin transform estimate

$$|\Re(s)| \leq 1/2 \implies \tilde{I}(s) \ll T^{1/4}(1 + |s|)^{-\infty}.$$

It follows readily that for fixed $\kappa > 0$, we incur the acceptable error $\mathrm{O}(T^{1/4-\kappa/2})$ by smoothly truncating the integral $\int_H \varphi$ to the range $\{T^{-\kappa} < |y| < T^\kappa\}$. If we seek only a qualitative subconvex bound, then we can take κ as small as we like, so there is no harm in truncating to $\{|y| = T^{o(1)}\}$. (Note that these arguments are specific to φ: for general vectors, one cannot always truncate to the range $|y| = T^{o(1)}$.)

The model problem is thus to bound

$$\int_{y \in H, y \asymp 1} \varphi(y)\, d^\times y. \tag{6.1}$$

7. Symmetries

We use a convolution kernel $\omega \in C_c^\infty(G)$ to "remember" many of the symmetries satisfied by φ. (When amplifying, we really take $\omega \in C_c^\infty(\mathrm{PGL}_2(\mathbb{A}))$.) Roughly speaking, we take ω to be a character multiple of an approximate subgroup of G:

$$\omega := \mathrm{vol}(J)^{-1} 1_J^{\mathrm{smooth}} \chi_\tau^{-1},$$

where:

- J is a subset of G roughly of the shape
$$J = (1 + \mathrm{O}(T^{-\varepsilon})) \cap (G_\tau + \mathrm{O}(T^{-1/2-\varepsilon})),$$
with G_τ the stabilizer of τ, as described in (5.3).
- 1_J^{smooth} is a smoothened characteristic function of J.
- χ_τ is the "approximate character" of J attached to τ, given near the identity in exponential coordinates by
$$\chi_\tau(\exp(X)) = e^{i\langle X, \tau\rangle}.$$

We may also describe ω in terms of the function $\mathfrak{g}^* \to \mathbb{C}$ obtained by taking the Fourier transform of the pullback $\omega \circ \exp$. This function is roughly a smoothened characteristic function of a "coin-shaped" neighborhood of τ, of thickness T^ε (resp. $T^{1/2+\varepsilon}$) in directions transverse (resp. tangential) to the coadjoint orbit $\mathcal{O}_\pi$ at τ. The intersection of this neighborhood with $\mathcal{O}_\pi$ has symplectic volume $\approx T^{2\varepsilon} \approx 1$. The orbit method heuristic suggests that for an irreducible representation σ of G, we have $\sigma(\omega) \approx 0$ unless σ is a principal series representation of parameter $T + \mathrm{O}(T^\varepsilon)$, in which case $\sigma(\omega)$ is approximately a projection onto a rank $\approx T^{2\varepsilon} \approx 1$ "subspace" of vectors microlocalized at τ. In particular,

$$\pi(\omega)\varphi \approx \varphi. \tag{7.1}$$

These heuristics and definitions can be made precise, and the above approximation holds in an extremely strong sense (i.e., up to $\mathrm{O}(T^{-\infty})$ with respect to any fixed seminorm).

For further informal discussion concerning ω in a general setting, see [13, §2]. For a precise discussion in the current rank one example, see §**??** of Quantum variance III.

8. The convexity threshold

Let's explain how to recover the convexity bound from here. Our task is to show that

$$\int_{y \in H, y \asymp 1} \pi(\omega)\varphi(y)\, d^\times y \ll T^{1/4}.$$

We view the square of the left hand side as one term arising from an integrated pretrace formula, like in the sup norm story [8]. Alternatively, we write the left hand side as the inner product of φ against a Poincaré series and apply Cauchy–Schwarz; see [13, §5.3]. Either way, we reduce to checking that

$$\int_{y_1, y_2 \asymp 1} \sum_{\gamma \in \Gamma} \omega(y_1^{-1} \gamma y_2) \ll T^{1/2}. \tag{8.1}$$

Since ω is supported on $1 + \mathrm{O}(T^{-\varepsilon})$, we see that the only γ that contribute are those in Γ_H. Combining the y_1 and y_2 integrals, it remains to check that

$$\int_{y \in H : y \asymp 1} \omega(y) \ll T^{1/2}. \tag{8.2}$$

To that end, we observe that

$$H \cap G_\tau = \{1\},$$

i.e., that no nontrivial matrices are simultaneously diagonal and of the form (5.3). (This is a baby case of the "stability" feature explained in [14, §1.9, §14].) It follows that (up to some ε's in the exponents)

$$H \cap J \subseteq \mathrm{O}(T^{-1/2}),$$

and so the volume of the integral in (8.2) is $\mathrm{O}(T^{-1/2})$.

On the other hand, the magnitude of the integrand is

$$\mathrm{vol}(J)^{-1} \approx T.$$

Indeed, J has dimensions roughly 1 along one direction and $T^{-1/2}$ along the remaining two directions.

These observations combine to give the required estimate (8.2).

Remark 5. It's clear in retrospect that we should have obtained such an estimate: the orbit method heuristic applied to ω suggests that the left hand side of (8.1) is a proxy for the sum of $|L(\pi, \frac{1}{2})|^2$ over T in an interval of width roughly $\mathrm{O}(T^\varepsilon)$ (see [13, §2.3]), which is of the appropriate size for an averaged Lindelöf estimate to recover convexity.

9. Amplification

This section is just a stub; see [13, §1.5, §2.7-2.10] for details and pictures relevant for this example. To carry out the amplification step, we basically need to know that the vector φ' obtained by averaging the translates of φ under elements of H of size $\asymp 1$ satisfies a matrix coefficient estimate $\langle g\varphi', \varphi' \rangle \ll T^{-\delta}$ except when g is very close to H. This is word-for-word what happens in the sup norm problem [8], where, if we replace H with $\mathrm{SO}(2)$, then φ' becomes something like the weight zero vector.

10. Comparison with Iwaniec

The "two arguments" are "equivalent." Recall that Iwaniec [7] starts with the approximate functional equation, applies Kuznetsov, and then applies Poisson summation to both variables. The reduction to (6.1) is essentially the approximate functional equation (TODO: explain this in some detail?). In the passage to (8.1), rather than applying the pretrace formula, we could have instead applied the Fourier expansion of φ and averaged each term in the resulting double sum using Kuznetsov. A couple applications of Poisson to the geometric side of Kuznetsov would then bring us right back to (8.1).

It's more interesting to compare the generalization of this argument to GL_3 with Blomer–Buttcane [3]. The arguments are again ultimately "equivalent," but the present approach seems less miraculous.

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